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#Assignment: 6B: K-Means Algorithm for two dimensional data items.
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#Date: 3/11/2026 Tuesday
#ESE Practical Exam- BA (NPTEL)
import numpy as np
import matplotlib.pyplot as plt

points = np.array([[1, 1], [1.5, 2], [3, 4], [5, 7], [3.5, 5],
                   [4.5, 5], [3.5, 4.5], [8, 8], [9, 9], [8.5, 9.5]])
k = 3
np.random.seed(42)
centroids = points[np.random.choice(len(points), k, replace=False)]
print("Initial centroids:\n", centroids)
```

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Initial centroids:
 [[9.  9. ]
 [1.5 2. ]
 [4.5 5. ]]
```

```
for it in range(1, 101):
    # distance of every point to every centroid (Euclidean)
    dist = np.linalg.norm(points[:, None] - centroids[None, :], axis=2)
    labels = dist.argmin(axis=1)
    new_centroids = np.array([
        points[labels == i].mean(axis=0) if np.any(labels == i) else centroids[i]
        for i in range(k)])
    print(f"\nIteration {it}: labels = {labels.tolist()}")
    if np.allclose(new_centroids, centroids):
        break
    centroids = new_centroids
```

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Iteration 1: labels = [1, 1, 2, 2, 2, 2, 2, 0, 0, 0]

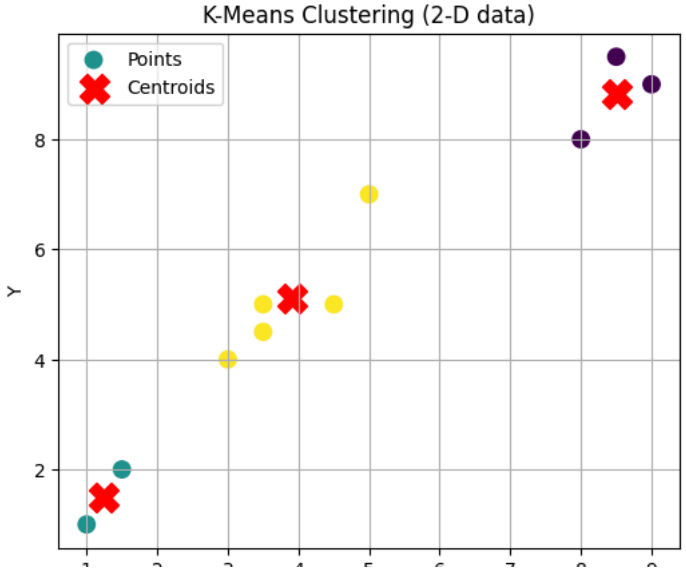
Iteration 2: labels = [1, 1, 2, 2, 2, 2, 2, 0, 0, 0]
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print("\nFinal centroids:\n", np.round(centroids, 2))
for i in range(k):
    print(f"Cluster {i+1}: {points[labels == i].tolist()}")
```

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Final centroids:
 [[8.5  8.83]
 [1.25 1.5 ]
 [3.9  5.1 ]]
Cluster 1: [[8.0, 8.0], [9.0, 9.0], [8.5, 9.5]]
Cluster 2: [[1.0, 1.0], [1.5, 2.0]]
Cluster 3: [[3.0, 4.0], [5.0, 7.0], [3.5, 5.0], [4.5, 5.0], [3.5, 4.5]]
```

```
plt.figure(figsize=(6, 5))
plt.scatter(points[:, 0], points[:, 1], c=labels, cmap='viridis', s=80, label='Points')
plt.scatter(centroids[:, 0], centroids[:, 1], c='red', marker='X', s=250, label='Centroids')
plt.title("K-Means Clustering (2-D data)")
plt.xlabel("X"); plt.ylabel("Y"); plt.legend(); plt.grid(True)
plt.show()
```


K-Means Clustering (2-D data)
Points
Centroids
Y
8
6
4
2